HARRY SILOAM SIDABALOK

Computer Science Student | Software Engineering

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Education

BINUS University **2024 – 2028**
Bachelor of Computer Science - Software Engineering GPA: 3.85/4.00 *Bekasi, Indonesia*

SMKN 1 Kota Bekasi **2021 – 2024**
Vocational High School - Software Engineering (RPL) *Bekasi, Indonesia*

Experience

Laboratory Assistant **2025 – 2026**
BINUS University *Jakarta, Indonesia*

- Actively engaged in training and mentoring students, guiding them through hands-on technical practices.
- Troubleshot complex coding issues and facilitated lab sessions to strengthen students' core programming and development skills.

Full Stack Developer Intern **2023 – 2024**
PT Agriva Teknologi Nusantara *Bekasi, Indonesia*

- Gained valuable hands-on experience designing and building interactive landing pages.
- Managed relational databases to ensure data integrity and fast retrieval times.

Projects

Edupose – AI Classroom Monitoring System *Python, YOLO, Raspberry Pi, Scikit-learn*

- Developed an AI-powered classroom monitoring system deployed on Raspberry Pi edge devices using RTSP video streaming.
- Leveraged YOLO pose estimation and custom keypoint feature engineering.
- Implemented ensemble machine learning classifiers to analyze student behavior and classroom engagement in real time.

Decentralized Web3 Chat Application *WebSocket, IPFS, Solidity, Relational DB*

- Built a real-time chat application utilizing WebSocket architecture.
- Integrated JWT authentication, smart contract verification, and decentralized storage via IPFS alongside a traditional relational database for persistent messaging.

TransTrack – Immutable Transaction Ledger *Solidity, Polygon, IPFS, PostgreSQL*

- Developed a transparent, immutable transaction tracking web application integrated with Solidity smart contracts on the Polygon Amoy testnet.
- Utilized IPFS (via Pinata) for decentralized file storage and PostgreSQL for rapid data retrieval.

Veritas – Fake News Detection Platform *Python, PyTorch, BERT, NLP*

- Created an AI-driven credibility analysis platform to detect fake news and misinformation.
- Leveraged PyTorch BERT transformers and TF-IDF Logistic Regression classifiers to evaluate text authenticity and context.

Technical Skills

Languages: Golang, Java, Python, SQL, JavaScript/TypeScript, HTML/CSS
Backend & Cloud: Spring Boot, Node.js, Docker, WebSockets, Tailscale, Cloudflare Tunnels, eBPF (Tetragon)
OS & Tools: Arch Linux, Git, KVM/libvirt, PostgresQL, IPFS